

837 to 835 End-to-End: Building It in Sterling Map Editor

A complete, hands-on lab for building this claim-to-remittance flow using only Sterling Map Editor and Sterling B2B Integrator — no ITX/WTX required

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Before You Begin

- Why this guide exists* Many EDI teams run their 834, 837, 835, 276/277, and 270 maps on IBM ITX (WebSphere Transformation Extender). This guide covers the same 837 claim-to-835-remittance scenario end to end, but builds it entirely with Sterling B2B Integrator's own Map Editor — for environments that don't have ITX installed, or for anyone who wants a native, no-extra-license way to see the whole flow work.
- Nothing about the business outcome changes. A claim arrives, gets translated, gets validated, goes to adjudication, and a remittance comes back — fully automated and fully tracked, start to finish. Only the tool that performs the two translations, and the way that tool plugs into the business process, is different from an ITX-based build.
- If you also use ITX* If your organization maintains an ITX-based version of this same flow, the next section — How Sterling Map Editor Compares to ITX — lines up every ITX concept against its Map Editor equivalent, so the two approaches are easy to compare side by side. If you've never touched ITX, skip straight to The Business Scenario; everything from that point on stands on its own.
- Who this guide is for* No prior Map Editor experience is assumed. A short glossary of Map Editor terms appears before the first hands-on exercise, and every exercise spells out exact menu paths, field values, and what you should see at each step.
- Scope* All data in this guide — provider name, payer name, patient name, member ID, NPI, and claim amounts — is fictional and used purely for illustration.

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How Sterling Map Editor Compares to ITX

Same job, different tool Map Editor and ITX solve the same problem — translating one structured document into another — and both are built around the same underlying idea: define what the source looks like, define what the target looks like, then describe field by field how one becomes the other. Where they differ is where that description lives, and how much of it is a fixed menu of choices versus free-form scripting.

Tooling and architecture

ITX concept	Map Editor equivalent	What's different
ITX Design Studio (an external design tool, needing a licensed WTX 8.4+ installation)	Sterling B2B Integrator Map Editor (a separate Windows download, obtained from within Sterling B2B Integrator itself)	No additional license. Map Editor is downloaded and installed once per machine that will build maps, from the same product you already run.
Type tree (built from a COBOL copybook, DFDL schema, or XML schema)	Map Editor type tree (built from an X12/EDIFACT/positional/XML/CSV standard chosen in the New Map Wizard, or an existing DDF file)	Same concept — a structural definition of the data — built from Sterling B2B Integrator's own standards database instead of an external schema.
A map service inside the business process that invokes the compiled map, requiring a licensed translation-engine runtime	No separate service. The compiled Map Editor map (.txo) is referenced by name directly inside the X12 envelope's Translation/Compliance Checking configuration	Architecturally simpler: translation happens as part of deenveloping (inbound) or enveloping (outbound) configuration, not as an extra invoked step.
A design tool's Test perspective (sample-input trace against the map)	Map Editor's Test Map function, or SI Map Test (tests the compiled .txo remotely before it is checked in)	Same purpose — feed sample input, inspect the output before deploying — different dialog.
Handled ad hoc, inside the map's function graph	GAP Analysis — an explicit, named phase in Map Editor's own map-building process: pair every output field with its input source before you link or rule anything	Map Editor makes this step formal; you'll use the same field-mapping-table format throughout this guide.

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*The five operations,
mechanism by
mechanism*

Operation	ITX mechanism	Map Editor mechanism	Note
Copy	A copy card or straight-through link in the map graph	Simple Link — drag the input field directly onto the compatible output field	Identical concept in both tools; this is the default, no-rule case.
Reformat	A reformat or string function built into the palette	Extended Rule — a short assignment statement in Map Editor's own scripting language, e.g. string concatenation with the + operator	Map Editor's fixed Standard Rule list has no generic "reformat" rule, so most reformatting becomes a small Extended Rule instead of a drag-and-drop function. This is the main asymmetry between the two tools.
Compute	A compute card (sum, arithmetic, comparison)	Use Accumulator Standard Rule for running sums/counts across a loop; Extended Rule for comparisons or multi-field arithmetic	Simple running totals map cleanly to Accumulators; anything conditional (like a MATCH/MISMATCH flag) needs an Extended Rule.
Default	A default value set on the target field	Use Constant Standard Rule	Same concept — the field always gets a fixed value when its own logic doesn't otherwise populate it.
Lookup	A lookup table (cross-reference table built into the map)	Use Code Standard Rule (an editable code list built inside Map Editor) or Use Select Standard Rule (a trading partner code list or process-data section)	Functionally identical — a maintained list of value pairs the map consults at runtime.

A bonus mechanism An ITX-based map typically captures the claim control number for tracking using the Correlation Service, configured on the surrounding business process. Map Editor can do the equivalent work inside the map itself: the Update Standard Rule writes a field's value straight into Sterling B2B Integrator's Correlation Data, with no separate business-process configuration needed. Exercise A below uses exactly this rule on the claim control number.

The Business Scenario

The provider Ridgeline Medical Group (fictional), NPI 1234567893, submits professional (837P) claims for office visits and diagnostic services.

The payer Meridian Health Plan (fictional), receiver ID 66783, adjudicates claims and returns 835 remittance advice showing what was paid, adjusted, and why.

The problem Ridgeline submits claims electronically and needs to know, without manual follow-up: that each claim arrived intact and passed compliance checking; that the claim was accepted (or rejected) at the business level; and eventually, exactly what was paid, what was adjusted, and why — all reconciled automatically against the original claim, with no manual re-keying at either end.

The shape of the solution This is the standard pattern for any request/response EDI pair — an inbound map and business process for one document type, paired with an outbound map and business process for the response. Here, the pairing is 837 in, 835 out; if you're used to ITX, only the translation engine changes.

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Solution Architecture: The End-to-End Flow

#	Stage	What happens
1	Provider submits 837P	Ridgeline's system sends an X12 837P interchange to Sterling B2B Integrator over AS2 or SFTP.
2	Inbound de-enveloping + translation	EDIDeenvelope.bp identifies the X12 interchange and starts X12DeenvelopeUnified.bp, which removes the ISA/GS/ST envelopes, performs compliance checking, and — because the ST/SE envelope's Translation/Compliance Checking configuration names it — translates the 837P using the Inbound 837 5010 Map you build in Map Editor.
3	Validate and route	A business process validates the mapped claim (the map's own ChargeReconciliationFlag, computed in Exercise A), then routes it to the adjudication system.
4	Adjudication (external)	The payer's adjudication system — outside Sterling B2B Integrator — decides what to pay, what to adjust, and why. It returns a Canonical Payment record.
5	Outbound translation + enveloping	The Outbound 835 5010 Map you build in Map Editor (Exercise C) translates the payment result into X12 835 segments; the EDI Encoder and EDI Enveloping services then apply the ST/GS/ISA envelopes via X12EnvelopeUnified.bp.
6	Delivery and acknowledgment	The enveloped 835 is delivered back to Ridgeline. Acknowledgments (TA1, 999, and separately 277CA for the original claim) flow in both directions along the way — this is native Sterling B2B Integrator behavior, independent of which tool did the translating.
7	Correlation and tracking	Because the Inbound 837 5010 Map's Update Standard Rule wrote the claim control number into Correlation Data, both the inbound 837 and the outbound 835 are searchable together by that one value at any time.

A note for ITX-based teams: Stages 1, 3, 4, 6, and 7 are native Sterling B2B Integrator behavior — they never depend on ITX in the first place. Only stage 2's and stage 5's translation mechanism change, and the change simplifies the architecture: there is no separate map-service step to configure, license, or troubleshoot.

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Environment & Prerequisites Checklist

- Map Editor installation*
- Map Editor is a separate Windows install, obtained from within Sterling B2B Integrator (the Download and Install section provides a Download Map Editor link) — install it once on the PC that will build and test maps.
 - Confirm the PC meets Map Editor's requirements: a supported Windows version, and — if you will work with DDF files — Microsoft XML Parser (MSXML) 4.0 installed on the same machine.
 - Confirm that PC has network access back to the Sterling B2B Integrator administrative interface, so maps can be checked in once built.

- Sterling B2B Integrator*
- No ITX, WTX, or any external translation-engine components are required anywhere in this design — that is the entire point of this approach.
 - The 837 (005010X222A1) and 835 (005010X221A1) transaction standards should already be present in Sterling B2B Integrator's standards database; Map Editor's New Map Wizard reads directly from that same standards database.
 - Trading partner profiles and a communication adapter for Ridgeline and Meridian (see the Trading Partner & Communication Setup section).

What you will build in this guide Two maps (Inbound 837 5010 Map, Outbound 835 5010 Map), wired into the native deenveloping/enveloping architecture described in Solution Architecture. Exercises A through D are written as literal, do-this-now lab steps you can follow at your own pace.

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Map Editor Concepts You'll Use in This Guide

Read this once before starting If this is your first time in Map Editor, read through the terms below once — every one of them appears in the exercises that follow, and knowing what they mean before you get there will save you from guessing.

Term	What it means
Map	The saved definition of how one document type translates into another — includes the input format, the output format, and every link and rule connecting them. Map Editor edits maps as .mxl source files.
Type tree	The structural definition of a document's fields, shown as an expandable tree on each side of the map window — one tree for the input format, one for the output format.
New Map Wizard	The guided dialog sequence (File > New) that creates a blank map: it asks what kind of map, names it, then walks you through defining the input and output formats.
Simple Link	The most basic connection between two fields: drag the input field onto the output field, and its value copies across unchanged. Used whenever no transformation is needed.
Standard Rule	One of a fixed set of built-in behaviors you attach to a field from its Properties dialog, Standard Rule tab. This guide uses Use Constant (a default value), Use Code (a lookup/code list), Use Accumulator (a running total), and Update (writing a value into Correlation Data).
Extended Rule	A short script you write yourself for anything a Simple Link or Standard Rule can't do — string concatenation, conditional logic, date reformatting, and multi-field arithmetic all typically need an Extended Rule.
GAP Analysis	The exercise of pairing every field in your output format with where its value comes from (or how it's derived) before you start linking fields or writing rules. The tables in this guide are the result of doing this in advance.
Compile	Turning your saved map (.mxl) into a translation object (.txo) that Sterling B2B Integrator can actually run. Map Editor does this from the File menu and shows a log of any errors.
Test Map / SI Map Test	Map Editor's built-in way to run sample input through your compiled map and inspect the output before you check the map in — this is how you catch mistakes early.
Check in	Publishing your compiled map into Sterling B2B Integrator so it can be selected by name elsewhere in the system — for example, from an envelope's configuration screen.
ST/SE, GS/GE, ISA/IEA	The three nested "envelopes" that wrap every X12 document: ST/SE wraps one transaction (a single 837 claim submission), GS/GE wraps a group of transactions, and ISA/IEA wraps the entire interchange sent in one connection.
Translation/Compliance Checking	The section of an inbound envelope's configuration where you tell Sterling B2B Integrator which map to run against an incoming document, and whether to check it for standards compliance first.

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Trading Partner & Communication Setup

<i>Basic trading partner profile</i>	Create a basic trading partner profile for your own organization (Meridian Health Plan, as the payer running Sterling B2B Integrator) and for Ridgeline Medical Group. If you communicate over AS2, create an AS2 trading profile instead of a basic one — AS2 requires certificate and MDN configuration that a basic profile does not carry.
<i>Communication adapter</i>	Configure the adapter that matches your agreement with the provider or clearinghouse: an AS2 adapter for AS2 exchange, or an SFTP Remote Profile if you exchange files over SFTP instead. An SFTP Remote Profile needs an SSH key pair, a Known Host entry, an Authorized User, a User Identity, and an SSH Host Identity — set these up under Trading Partner > Advanced Management > SSH before configuring the adapter itself.
<i>Two adapters, two directions</i>	You need one adapter path to collect the inbound 837 and pass it to the deenveloping process, and a second to deliver the outbound 835 once it's enveloped. Trading partner and communication setup is entirely independent of which tool performs the translation.

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Inbound Side: Receiving the 837P Claim

Sample inbound 837P (fictional) A simplified, realistic 837P interchange — one claim, one patient, two service lines — used throughout this guide:

```
ISA*00*                *00*                *ZZ*RIDGELINEMED    *ZZ*MERIDIANHP    *260115*0930*^^*00501*000012345*0*P*:~
GS*HC*RIDGELINEMED*MERIDIANHP*20260115*0930*54321*X*005010X222A1~
ST*837*0001*005010X222A1~
BHT*0019*00*0001*20260115*0930*CH~
NM1*41*2*RIDGELINE MEDICAL GROUP*****46*1234567893~
NM1*40*2*MERIDIAN HEALTH PLAN*****46*66783~
HL*1**20*1~
NM1*85*2*RIDGELINE MEDICAL GROUP*****XX*1234567893~
HL*2*1*22*0~
SBR*P*18*****CI~
NM1*IL*1*SAMPLE*JOHN*Q***MI*M00000001~
CLM*ACCT100425*185.00***11:B:1*Y*A*Y*Y~
HI*ABK:M25569~
LX*1~
SV1*HC:99213*95.00*UN*1***1~
DTP*472*D8*20260114~
LX*2~
SV1*HC:71046*90.00*UN*1***1~
DTP*472*D8*20260114~
SE*17*0001~
GE*1*54321~
IEA*1*000012345~
```

Reading the key segments

Segment	What it carries in this claim
ISA / GS / ST	Interchange, group, and transaction-set envelopes — removed by the deenveloping process before mapping.
BHT	Beginning of Hierarchical Transaction — BHT02 = 00 (Original), BHT06 = CH (Chargeable claim, not an encounter).
NM1*41 / NM1*40	Submitter (Ridgeline) and Receiver (Meridian) — loops 1000A/1000B.
HL / NM1*85	Billing Provider hierarchical level and name/NPI — loop 2000A/2010AA.
HL / SBR / NM1*IL	Subscriber hierarchical level, payer-responsibility and relationship code, and subscriber name/member ID — loop 2000B/2010BA.
CLM	Claim Information (loop 2300) — CLM01 = patient control number, CLM02 = total charge, CLM05 = place of service / facility qualifier / claim frequency.
HI	Health Care Diagnosis Codes — qualifier ABK = principal ICD-10 diagnosis.
LX / SV1 / DTP	Service line number, Professional Service (procedure code, charge), and service date — loop 2400, repeated once per line.
SE / GE / IEA	Transaction, group, and interchange trailers.

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Exercise A: Building the Inbound Map in Sterling Map Editor

Target: Canonical What your adjudication system will consume once the Inbound 837 5010 Map runs:
Claim record

HEALTHCLAIM

```
ClaimControlNumber ..... ACCT100425
BillingProviderNPI ..... 1234567893
BillingProviderName ..... Ridgeline Medical Group
SubscriberMemberID ..... M00000001
SubscriberName ..... Sample, John Q
SubscriberRelationshipCode . 18
PlaceOfServiceCode ..... 11
PrincipalDiagnosis ..... M25.569
TotalChargeSubmitted ..... 185.00
SERVICELINES (group, repeats)
  LineNumber ..... 1
  ProcedureCode ..... 99213
  Charge ..... 95.00
  ServiceDate ..... 2026-01-14
ComputedLineTotal ..... 185.00
ChargeReconciliationFlag ... MATCH
```

Why a positional layout here *This internal format has no predefined industry standard behind it, so it's normally built as a Positional layout (a fixed, self-defined record structure) in Map Editor's New Map Wizard. If your adjudication system specifically requires XML, Map Editor's New Map Wizard also offers XML as an output format choice — the GAP Analysis, links, and rules below work the same way either way.*

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*GAP Analysis: 837P to
Canonical Claim*

Input field (837P)	Output field (Canonical Claim)	Link type	Map Editor mechanism
CLM01	ClaimControlNumber	Simple Link + Standard Rule	Drag CLM01 to ClaimControlNumber, then add an Update Standard Rule (table = Correlation Data, correlation parameter name = Claim Control Number) so the claim becomes searchable.
NM1*85, element 09	BillingProviderNPI	Simple Link	Drag directly — both are compatible string fields.
NM1*85, element 03	BillingProviderName	Simple Link	Drag directly.
NM1*IL, elements 03 + 04	SubscriberName	Extended Rule	#SubscriberName = \$Claim.#NM1_IL_03 + ", " + \$Claim.#NM1_IL_04; — the + operator performs concatenation in Map Editor's extended-rule language.
NM1*IL, element 09	SubscriberMemberID	Simple Link	Drag directly.
SBR01	SubscriberRelationshipCode	Use Constant Standard Rule	If SBR01 arrives blank, the Use Constant Standard Rule supplies the default "18" (self) so the field is never empty.
CLM05, first component	PlaceOfServiceCode	Use Code Standard Rule	Build a code list mapping X12 place-of-service values to your internal codes (here, a 1:1 pass-through for "11"); the code list also lets you Raise a compliance error if an unexpected value arrives.
HI segment, ABK element	PrincipalDiagnosis	Extended Rule	Inserts the decimal point that X12 omits (M25569 to M25.569) using Map Editor's Extended Rule function library — check Help > Extended Rule Functions in your version for the exact function name.
DTP*472, each service line	ServiceLines/ServiceDate	Extended Rule	Reformats CCYYMMDD to CCYY-MM-DD, once per loop occurrence.
SV1 charge, every LX/SV1 loop	ComputedLineTotal	Use Accumulator Standard Rule	Accumulator accum(1) adds each line's SV1 charge as the loop repeats; after the loop, #ComputedLineTotal = accum(1);.
ComputedLineTotal vs. CLM02	ChargeReconciliationFlag	Extended Rule	See the extended rule below — this is the one field with real conditional logic.

The one Extended Rule worth reading closely ChargeReconciliationFlag is the highest-value field in this map: a claim where the service-line charges don't sum to the claim-level total is either malformed or represents a billing error, and catching it here — before adjudication — is far cheaper than catching it after a payment has already been calculated against the wrong number.

```
IF #ComputedLineTotal = $Claim.#CLM02 THEN
    #ChargeReconciliationFlag = "MATCH";
ELSE
    #ChargeReconciliationFlag = "MISMATCH";
```

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Exercise A: Step by Step — Build the Inbound Map *(90-120 min, first time through)*

Instructions

1. Launch Sterling B2B Integrator Map Editor. If you haven't installed it yet, log in to the Sterling B2B Integrator web console, find the Map Editor download link, download the Windows installer, and run it — this only needs to be done once per machine.
2. From the Map Editor main menu, click File, then click New. This opens the New Map Wizard, a short series of dialog boxes that asks a few questions before creating a blank map.
3. On the first wizard screen, set "What kind of map are you creating?" to Sterling B2B Integrator, "What is the name of the map?" to Inbound 837 5010 Map, and "What is your name" to your own login name. Click Next.
4. The wizard now asks for the Input Format — the shape of the data coming into the map (the 837P claim). Select "Create a new data format using this syntax," choose "Delimited EDI (ANSI X12, UN EDIFACT, Tradacoms, etc.)" from the list, and click Customize.
5. In the New Delimited EDI Wizard that opens, leave "Import Code List" selected and click Next. On the following screen, choose EDI from the ODBC data source list and click Next again.
6. Set the three dropdowns that appear — standards agency, version, and transaction set — to X12, 005010X222A1, and 837 (Health Care Claim). Click Next, then Finish. You're returned to the New Map Wizard with the input side already defined.
7. Click Next to reach the Output Format screen — this defines the shape of the data leaving the map, your internal Canonical Claim record. Since there's no industry standard for this internal format, select "Create a new data format using this syntax" and choose Positional. (If your adjudication system consumes XML instead, choose XML here — the rest of this exercise works the same way.)
8. Click Next, then Finish. Map Editor opens showing two panels side by side: the 837P structure on the left, already populated from the X12 standard, and a blank output tree on the right, ready for you to build.
9. Build the output tree to match the Canonical Claim layout shown above. Right-click the root of the output tree, choose Insert > Group, name it SERVICELINES, and mark it as repeating — it will hold one entry per service line. For every other field in the layout (ClaimControlNumber, BillingProviderNPI, BillingProviderName, SubscriberMemberID, SubscriberName, SubscriberRelationshipCode, PlaceOfServiceCode, PrincipalDiagnosis, TotalChargeSubmitted, ComputedLineTotal, and ChargeReconciliationFlag), right-click the appropriate parent, choose Create Sub > Record, then Insert > Field, and type the field name exactly as shown. Save your work often as you go (File > Save, or Ctrl+S).
10. With both sides of the map built, work through the GAP Analysis table above field by field. For each output field, the table already tells you whether it needs a Simple Link, a Standard Rule, or an Extended Rule — use it as your build checklist for the remaining steps.
11. Create the Simple Links first, since they're the quickest to do: click and hold on CLM01 in the left (input) tree, drag it onto ClaimControlNumber in the right (output) tree, and release — a connecting line should appear. Repeat the same drag-and-drop for NM1*85 element 03 onto BillingProviderName, NM1*85 element 09 onto BillingProviderNPI, and NM1*IL element 09 onto SubscriberMemberID.
12. Add the Update Standard Rule: right-click ClaimControlNumber (already linked above) and choose Properties, then click the Standard Rule tab. Set "Please select the standard rule to use" to Update, "Please select the table (or group) to update" to Correlation Data, "Enter the correlation parameter name" to Claim Control Number, and "Please select the column (or field) to update" to Correlation Value. Click OK.
13. Add the Use Constant Standard Rule: right-click SubscriberRelationshipCode, choose Properties > Standard Rule tab, select Use Constant, and enter 18 as the default value. Click OK.
14. Add the Use Code Standard Rule: right-click PlaceOfServiceCode, choose Properties > Standard Rule tab, select Use Code, click Edit to create a new code list, and add an entry mapping the X12

value 11 to your internal value 11 (a direct pass-through for this example). Click OK to save the code list, then OK again to close the dialog.

15. Add the Use Accumulator Standard Rule: right-click ComputedLineTotal, choose Properties > Standard Rule tab, select Use Accumulator, and point it at accumulator #1. This accumulator adds up every service line's charge as the SV1 loop repeats.
16. Add the remaining Extended Rules. Right-click SubscriberName and choose Extended Rule; in the editor that opens, type the concatenation shown in the GAP Analysis table, then click OK. Repeat the same process for PrincipalDiagnosis (inserting the decimal point) and for the DTP*472 date reformat, using the GAP Analysis table for what each one needs to do.
17. Add the last Extended Rule: right-click ChargeReconciliationFlag, choose Extended Rule, and type the four lines shown above (the IF/THEN/ELSE comparing ComputedLineTotal to CLM02). Click OK.
18. Save the map (File > Save As, choosing a folder you'll remember), then click Compile from the File menu to turn it into a translation object. Review the compile log Map Editor displays — a clean compile shows zero errors; if you see errors, they'll point you to the exact field or rule to fix.
19. Test the map before checking it in: from the Functions menu, choose SI Map Test (or Map Editor's own Test Map option), point it at the sample 837P shown earlier as input, run the test, and compare the output against the Canonical Claim target shown above — pay particular attention to ChargeReconciliationFlag, which should read MATCH.
20. Once the test output matches, check the map in to Sterling B2B Integrator (from Map Editor's administration options, or the web console's map administration area) as Inbound 837 5010 Map. This makes it available to select from the envelope configuration in Exercise B.

Exercise B: Wiring the Inbound Map into the Business Process

EDIDeenvelope.bp The predefined entry point for every inbound EDI document. It identifies the interchange type in the incoming document and starts the appropriate standard-specific subprocess — for X12, that is X12DeenvelopeUnified.bp.

X12DeenvelopeUnified.bp Removes the ISA/GS/ST envelopes and performs compliance checking. This is where the map you just built gets invoked — not through a separate business-process step, but through the envelope's own configuration, which you'll set up in this exercise.

Service (inside X12DeenvelopeUnified.bp)	What it does
X12 Deenvelope	Removes X12 envelopes and performs compliance checking (not sequence checking). Also generates TA1 acknowledgments and, when configured, the 999 functional acknowledgment for the interchange.
X12 TA1 Reconciliation	Reconciles interchange-level (TA1) acknowledgments sent back by a trading partner.
EDI Post Processor	Handles deferred sequence checking and acknowledgment processing in parallel with translation, when control-number sequence checking is required.

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Exercise B: Step by Step — Wire the Inbound Map into the Envelope *(30-45 min)*

Instructions

1. In the Sterling B2B Integrator web console, go to Trading Partner > Document Envelopes and open the Inbound X12 ST/SE Envelope for the 837 transaction between Ridgeline and Meridian. If it doesn't exist yet, create a new one and step through its setup wizard, accepting the defaults on any screen not mentioned below.
2. Navigate to the envelope's Translation/Compliance Checking page. Set "Translate and/or Compliance Check Document" to Yes.
3. Continue to the Map Name settings. Set "Map Name Mode" to Specify.
4. In the field that appears, set "Compliance Check Map Name" to Inbound 837 5010 Map — the exact map you checked in at the end of Exercise A. If it doesn't appear in the list, double-check that it was checked in successfully before continuing.
5. Continue to the Compliance Checking page. Set "Translate transactions" to Yes. For "Handling of non-compliant Transactions," choose "Accept compliant and non-compliant transactions" while you're testing — you can tighten this later once you're confident the map is solid.
6. Save the envelope. Then confirm the inbound Group (GS/GE) and Interchange (ISA/IEA) envelopes for Ridgeline and Meridian already exist and reference this ST/SE envelope — every X12 document needs all three envelope levels configured, regardless of which tool performs the mapping.
7. Test end to end: submit the sample 837P shown earlier through your inbound communication adapter, or use Business Process > Manager > Execution Manager to run it manually against a saved copy of the sample file. Watch the business process execute in Business Process Visibility, and confirm it completes without a compliance error.
8. Open the completed business process instance and inspect its process data to confirm the Canonical Claim record it produced matches the target shown in Exercise A, including ChargeReconciliationFlag = MATCH. Notice that translation happened automatically as part of deenveloping — there is no separate map-invocation step anywhere in the process.

Adjudication: What Happens Between the Two Maps

Outside Sterling B2B Integrator Adjudication — deciding what to pay, deny, or adjust, and why — is the payer's core business logic, not something Sterling B2B Integrator or Map Editor performs. Sterling B2B Integrator's job ends when it hands the adjudication system a clean, validated Canonical Claim record, and picks back up when that system returns a result.

In this example, assume the adjudication system applies Ridgeline's contracted fee schedule, reduces each line by a contractual adjustment, and returns a Canonical Payment record with the result. That result is the input to the outbound map, below.

*Sample Canonical
Payment record
(fictional result)*

CLAIMPAYMENT

ClaimControlNumber ACCT100425
PayerClaimControlNumber 66783-0001
ClaimStatusCode PAID_PRIMARY
TotalChargeSubmitted 185.00
TotalPaid 148.00
PatientResponsibility 37.00
SERVICELINES (group, repeats)
 ProcedureCode 99213
 Charged 95.00
 Paid 76.00
 AdjustmentReasonCode FEE_SCHEDULE
 AdjustmentAmount 19.00
PaymentMethod ACH
PaymentDate 2026-01-22

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Outbound Side: Generating the 835 Remittance

Sample outbound 835 (fictional) What the map and enveloping process produce from the Canonical Payment record above:

ISA*00* *00* *ZZ*MERIDIANHP *ZZ*RIDGELINEMED *260122*1200**^*00501*000098765*0*P*:~
GS*HP*MERIDIANHP*RIDGELINEMED*20260122*1200*77001*X*005010X221A1~
ST*835*0001*005010X221A1~
BPR*I*148.00*C*ACH*CTX*01*999999999*DA*123456789**01*888888888*DA*000111222*20260122~
TRN*1*66783000199*199999999~
N1*PR*MERIDIAN HEALTH PLAN~
N1*PE*RIDGELINE MEDICAL GROUP*XX*1234567893~
CLP*ACCT100425*1*185.00*148.00**12*66783-0001*11~
CAS*PR*1*37.00~
NM1*QC*1*SAMPLE*JOHN*Q~
SVC*HC:99213*95.00*76.00**1~
CAS*CO*45*19.00~
DTM*472*20260114~
SVC*HC:71046*90.00*72.00**1~
CAS*CO*45*18.00~
DTM*472*20260114~
SE*16*0001~
GE*1*77001~
IEA*1*000098765~

*Reading the key
segments*

Segment	What it carries in this remittance
BPR	Financial Information — BPR02 = total payment amount, BPR04 = payment method (ACH here).
TRN	Reassociation Trace Number — ties this remittance back to the actual payment for automated reconciliation.
N1*PR / N1*PE	Payer (Meridian) and Payee (Ridgeline) identification — loops 1000A/1000B.
CLP	Claim Payment Information (loop 2100) — status code, total charge, total paid, payer's own claim control number.
CAS*PR	Claim-level adjustment, group code PR (Patient Responsibility) — here, the patient's share.
SVC / CAS*CO	Service Payment Information per line, and a service-level adjustment with group code CO (Contractual Obligation) — reason code 45 (charge exceeds fee schedule).
DTM*472	Service date at the line level, echoing the original claim's service date.

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Exercise C: Building the Outbound Map in Sterling Map Editor

*GAP Analysis:
Canonical Payment to
835*

Input field (Canonical Payment)	Output field (835)	Link type	Map Editor mechanism
ClaimControlNumber	CLP01	Simple Link	Drag directly — both are compatible string fields.
ClaimStatusCode	CLP02	Use Code Standard Rule	Code list maps internal status (e.g. PAID_PRIMARY) to the X12 CLP02 code (e.g. 1).
TotalChargeSubmitted / TotalPaid	CLP03 / CLP04	Simple Link	Drag directly, one per field.
TotalPaid, summed across every claim in the batch	BPR02	Use Accumulator Standard Rule	Accumulator adds every claim's TotalPaid as the batch loop repeats; BPR02 reads the final accumulator value.
PatientResponsibility	CAS*PR amount	Simple Link	Drag directly, tagged with group code PR by a Use Constant Standard Rule on the adjacent code field.
AdjustmentReasonCode	CAS02 (per line)	Use Code Standard Rule	Code list maps internal reason (e.g. FEE_SCHEDULE) to the X12 CARC code (e.g. 45).
AdjustmentAmount (per line)	CAS03 (per line)	Simple Link	Drag directly.
PaymentDate	BPR16	Extended Rule	Reformats CCYY-MM-DD to CCYYMMDD — the mirror image of the inbound DTP*472 reformat in Exercise A.
PaymentMethod, if blank	BPR04 fallback	Use Constant Standard Rule	If PaymentMethod arrives blank, defaults to "CHK" (check).

Same five operations, running in reverse *This map uses exactly the same operations as the inbound map — Simple Link, Use Code, Use Accumulator, Use Constant, Extended Rule — just running the other direction: internal codes are looked up out to X12 codes instead of in, and BPR16's date reformat is the mirror image of the inbound DTP*472 reformat. The operations don't change between directions, only the mechanism that implements each one.*

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Exercise C: Step by Step — Build the Outbound Map *(60-90 min)*

Instructions

1. Open Map Editor and select File > New to start a fresh map.
2. On the first wizard screen, set the map name to Outbound 835 5010 Map, provide your name, and click Next.
3. On the Input Format screen, select "Create a new data format using this syntax" and choose Positional (or XML, matching whatever you chose for the Canonical Claim record in Exercise A). Click Next.
4. Build the input tree to match the Canonical Payment layout shown earlier — using the same right-click Insert > Group / Create Sub > Record / Insert > Field technique from Exercise A — for ClaimControlNumber, PayerClaimControlNumber, ClaimStatusCode, TotalChargeSubmitted, TotalPaid, PatientResponsibility, a repeating SERVICELINES group (ProcedureCode, Charged, Paid, AdjustmentReasonCode, AdjustmentAmount), PaymentMethod, and PaymentDate. Click Next once the tree is built, then Finish.
5. Click Next again to reach the Output Format screen. Select "Create a new data format using this syntax," choose Delimited EDI, and click Customize.
6. In the New Delimited EDI Wizard, accept the defaults through the ODBC screen (as in Exercise A), then set standards agency to X12, version to 005010X221A1, and transaction set to 835 (Health Care Claim Payment/Advice). Click Next, then Finish, then Finish again to complete the New Map Wizard.
7. You now have the Canonical Payment structure on the left and the 835's X12 structure — already populated from the standard — on the right.
8. Work through the GAP Analysis table above, field by field, exactly as you did in Exercise A.
9. Create the Simple Links: drag ClaimControlNumber onto CLP01, TotalChargeSubmitted onto CLP03, TotalPaid onto CLP04, PatientResponsibility onto the CAS*PR amount field, and each line's AdjustmentAmount onto CAS03.
10. Add the Standard Rules: right-click CLP02 and set Use Code with a new code list mapping ClaimStatusCode values (e.g. PAID PRIMARY) to X12 codes (e.g. 1); right-click CAS02 and set Use Code with a code list mapping AdjustmentReasonCode values (e.g. FEE_SCHEDULE) to CARC codes (e.g. 45); right-click BPR02 and set Use Accumulator to total every claim's TotalPaid in the batch; right-click the BPR04 fallback and set Use Constant with default value CHK.
11. Add the Extended Rule on BPR16 that reformats PaymentDate from CCYY-MM-DD to CCYYMMDD.
12. Save the map, then compile it and review the compile log for errors.
13. Test the map against the sample Canonical Payment record shown earlier, and confirm the output matches the sample 835 shown, including the CLP02, CAS, and SVC values.
14. Check the map in to Sterling B2B Integrator as Outbound 835 5010 Map, so it's available to select in Exercise D.

Exercise D: Wiring the Outbound Map into the Business Process

Service / process	What it does
EDI Encoder Service	Determines which transaction-level (ST/SE) envelope applies and whether translation is needed, using three required parameters: AcceptorLookupAlias, SenderID, and ReceiverID.
EDI Enveloping Service	Determines the correct predefined enveloping business process to run — here, X12EnvelopeUnified — and either runs it immediately or on a deferred schedule.
X12EnvelopeUnified.bp	Applies the ST/SE transaction envelope, then wraps it in the GS/GE group envelope, then the ISA/IEA interchange envelope, using the envelope data to locate the trading partner and deliver the document.

Exercise D: Step by Step — Wire the Outbound Map into the Envelope *(30-45 min)*

Instructions

1. In the Sterling B2B Integrator web console, go to Trading Partner > Document Envelopes and open (or create) the Outbound X12 ST/SE Envelope for the 835 transaction between Meridian and Ridgeline.
2. Navigate to the envelope's Map Name settings and set "Map Name Mode" to Specify.
3. Set "Map Name" to Outbound 835 5010 Map — the map you checked in during Exercise C.
4. Build the outbound business process: an EDI Encoder Service step (AcceptorLookupAlias, SenderID = Meridian, ReceiverID = Ridgeline, set directly or resolved via XPath), followed by an EDI Enveloping Service step in Immediate mode.
5. Add an Invoke Business Process step naming the process the EDI Enveloping Service returns, so X12EnvelopeUnified.bp actually runs.
6. Configure the outbound communication adapter (AS2 or SFTP) to deliver the enveloped 835 to Ridgeline.
7. Test end to end: feed the sample Canonical Payment record through the outbound process and confirm the delivered 835 matches the sample shown above.

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Acknowledgments & Compliance Checking

Three layers of acknowledgment This is native Sterling B2B Integrator behavior that has nothing to do with which tool built the translation map. For every 837 Ridgeline sends, up to three separate acknowledgments can come back — each catching a different class of problem, at a different level:

Layer	Acknowledgment	What it confirms
Interchange	TA1	The ISA/IEA envelope itself is well-formed. Generated only when there's an interchange-level problem.
Syntax / TR3 compliance	999 (Implementation Acknowledgment)	The transaction is syntactically valid and complies with the 005010X222A1 implementation guide. This is a technical, EDI-team-level check — the 999 is the current HIPAA 5010 standard for this.
Business / claim level	277CA (Claim Acknowledgment)	Each individual claim was accepted or rejected against the payer's business rules — generated only after the 999 accepts the interchange, and separately for each claim. An accepted claim's 277CA carries the payer's own claim control number, useful later for 276/277 status inquiries; a rejected claim's 277CA carries the specific business error.

Why this matters for your team A 999 rejection means the EDI/software team has a syntax or format problem to fix. A 277CA rejection means the billing team has a data problem — a bad member ID, an invalid procedure code — that a resubmission needs to correct. Routing these to the right team, rather than treating every rejection the same way, is one of the most common real-world sources of delay in claim processing.

Control numbers: preventing double-processing Sterling B2B Integrator's Control Numbers apply directly here. Sequence checking confirms Ridgeline's interchanges, groups, and transaction sets arrive in order — waiting up to 30 minutes (SequenceCheckRetryTimeout, configurable in envelope.properties) for a missing one before erroring. Duplicate checking, tracked for a configurable number of days, is what stops the same 837 claim from being processed — and paid — twice if it's ever resubmitted by mistake.

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Tracking Claims: Correlation and Overdue Acknowledgment

<i>The Correlation Service, plus a head start from the map itself</i>	The Correlation Service runs automatically whenever the EDI Enveloping, EDI Overdue Acknowledgment Status, or EDI Deenveloping services are used — no setup required. It collects Sender ID, Receiver ID, Control Number, Acknowledgment Requested/Status, Standard, Functional ID, Version, envelope date/time, and Transaction Set ID for every document, at every enveloping and de-enveloping stage. This guide's map adds one more searchable value for free: the Update Standard Rule you configured on ClaimControlNumber in Exercise A wrote that value directly into Correlation Data as it mapped, so a search on EDI Correlation with Name = Claim Control Number, Value = ACCT100425 will find every document tied to this specific claim — inbound and outbound alike.
<i>Overdue Acknowledgment</i>	The predefined OverdueAckCheck business process finds outbound interchanges and groups whose expected inbound acknowledgment hasn't arrived. For the 835 you just sent Ridgeline, this means: specify in the outbound envelope that an acknowledgment is expected, run OverdueAckCheck on a recurring schedule using Sterling B2B Integrator's own scheduling feature (Business Process > Scheduling), and review the report it produces — blank means nothing is overdue.

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Best Practices Applied

- Assign Service* Where the map itself doesn't compute a field — for example, any additional business logic your business process applies after translation — use one Assign Service rather than several separate Assign activities. Consolidating into a single Assign Service typically yields a significant, measurable performance improvement for exactly this pattern.
- Invoke Sub-Process mode* Route the validated Canonical Claim record to your adjudication system using Sync mode, not Async — you need to know immediately if adjudication fails, and this is a single claim needing an ordered result, not a fire-and-forget batch job.
- message_to_child / message_to_parent* An 837 with dozens or hundreds of service lines carries a large amount of process data by the time it's been de-enveloped and mapped. Before invoking the adjudication sub-process, create a `message_to_child` tag in the parent holding only the Canonical Claim record the sub-process actually needs — not the original raw 837 segments. This is a well-known performance consideration for large transactions, and it matters more here than in almost any other document type your team handles, given how large a single claim's process data can get.
- Release Service* Once the map has produced the Canonical Claim record and the reconciliation check has passed, release the raw de-enveloped 837 segments from process data before routing to adjudication — they're no longer needed, and holding them adds no value while consuming memory on every one of potentially thousands of daily claims.

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End-to-End Verification Checklist

1. Inbound 837 5010 Map compiles cleanly and passes Test Map / SI Map Test against the sample 837P.
2. The map produces a Canonical Claim record exactly matching the target shown, including ChargeReconciliationFlag = MATCH.
3. Deliberately break the sample (change one SV1 charge without updating CLM02) and confirm ChargeReconciliationFlag correctly flips to MISMATCH.
4. Sample 837P submitted through the inbound adapter is de-enveloped, translated, and compliance-checked with no separate map-invocation step anywhere in the business process.
5. A 999 is generated for the inbound interchange, and a 277CA is generated separately at the claim level.
6. Outbound 835 5010 Map compiles cleanly and produces an 835 matching the sample shown, including the correct CLP02, CAS, and SVC values.
7. The outbound 835 is successfully enveloped by X12EnvelopeUnified.bp and delivered via the configured adapter.
8. EDI Correlation search for Claim Control Number = ACCT100425 returns both the inbound 837 and the outbound 835, without any extra Correlation Service configuration beyond the map's own Update Standard Rule.
9. Resubmitting the identical sample 837P a second time is caught by duplicate control-number checking, not processed (and paid) again.

What's next Once this pair is working end to end, the same architecture — inbound map + envelope-level translation, outbound map + envelope-level translation, acknowledgment handling, correlation tracking — applies directly to 834, 276/277, and 270 maps, whether they stay on ITX or move to Map Editor. 276/277 is the closest parallel to build next: 276 (inquiry) inbound mirrors this guide's 837 side, and 277 (response) outbound mirrors this guide's 835 side, almost segment-family for segment-family.

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Appendix: What Changes If You're Coming From ITX

At a glance

Component	ITX-based approach	Map Editor approach
Design tool	An external design studio, needing a licensed translation-engine runtime	Sterling B2B Integrator Map Editor (included, separate Windows install)
Runtime wiring	A map-service business-process step, needing a licensed runtime	No separate service — compiled map (.txo) referenced by name in the envelope's Translation/Compliance Checking (inbound) or Map Name (outbound) configuration
Copy	A copy card	Simple Link
Reformat	A reformat function	Extended Rule (assignment statement)
Compute	A compute card	Use Accumulator Standard Rule, or Extended Rule
Default	A default value on the target field	Use Constant Standard Rule
Lookup	A lookup table card	Use Code Standard Rule or Use Select Standard Rule
Correlation capture	A business-process Assign step after the map	Update Standard Rule inside the map itself
Testing	The design studio's own test perspective	Map Editor Test Map / SI Map Test
Everything else (envelopes, deenveloping, enveloping, TA1/999/277CA, control numbers, Correlation Service, Overdue Acknowledgment, best practices)	Identical	Identical — none of it depends on the translation tool

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Fictional elements *Ridgeline Medical Group, Meridian Health Plan, the patient "John Q. Sample," and all NPIs, member IDs, control numbers, and dollar amounts in this guide are invented for illustration and do not correspond to any real organization, provider, or individual.*

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